Chapter 6
Complex Integration

Line integral in the complex plane

Complex definite integrals are called (complex) **line integrals**. They are written as

$$\int_C f(z)dz.$$

Here the **integrand** $f(z)$ is integrated over a given curve *C.* This curve *C* in the complex plane is called the **path of integration**.

If *C* is a **closed path** (one whose terminal point coincides with its initial point), then it is denoted by

$$\oint_C f(z)dz.$$

Partitioning of path *C*: If *C* is a combination of C_1 and C_2 then,

$$\int_C f(z)dz = \int_{C1} f(z)dz + \int_{C2} f(z)dz.$$

We may represent C by a parametric representation $z(t) = x(t) + iy(t),\ a \le t \le b$. That is,

$$\int_C f(z)dz = \int_C f(z(t))z'(t)dt.$$

The sense of increasing *t* is called the **positive sense** on *C.*

Note: Parametric representation of any curve is not unique.

Example 1: Sketch path C consisting of two-line segments, one from *z = 0* to *z = 2* and other from *z = 2* to *z = 3+i* , hence evaluate $\int_C f(z)dz$, if *f(z)* = z^2.

Solution:

Given, C consists of two-line segments, one from *z = 0* to *z = 2* and other from *z = 2* to *z = 3+i.*

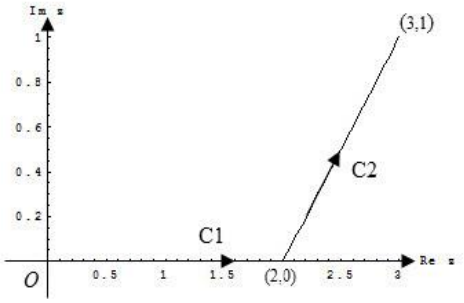


Fig: 5

Along C1:

Equation of the line, which passes through (0,0) and (2,0), is $y = 0$

$f(z) = z^2 = (x + iy)^2 = x^2$

[using $y = 0$]

We know, $z = x + iy = x$, $dz = dx$

and x varies from 0 to 2

$\int_{C1} f(z)dz = \int_0^2 x^2 dx = \frac{8}{3}$.

Along C2:

Equation of the line, which passes through (2,0) and (3,1) is,

$y - 0 = \left(\frac{1-0}{3-2}\right)(x-2) \Rightarrow y = x - 2.$

$f(z) = z^2 = (x + iy)^2 = [(y + 2) + iy]^2$ [using $x = y + 2$]

We know, $z = x + iy = y + 2 + iy,\ dz = (1 + i)dy$ and y varies from 0 to 1.

$\int_{C2} f(z)dz = \int_0^1 [(y+2)+iy]^2(1+i)\,dy = i\int_0^1 (4+4i-2y^2+2iy^2+8iy)dy = \frac{10}{3} + \frac{26}{3}i$

Now, $\int_C f(z)\,dz = \int_{C1} f(z)dz + \int_{C2} f(z)dz = 6 + \frac{26}{3}i.$

Example 8: Sketch path C from $z=-1$ to $z=1$ along the upper half of the circle $|z|=1$ and hence evaluate $\int_C f(z)dz$, where $f(z)=\bar{z}$.

Solution:

Given, C is the upper half of the circle $|z|=1$ from z = -1 to z = 1.

$|z|=1, z=1.e^{i\theta}, dz = ie^{i\theta}d\theta$,

where

θ varies from π to 0, and $f(z)=\bar{z}= e^{-i\theta}$

Now, $\int_C f(z)dz = \int_\pi^0 e^{-i\theta}\, ie^{i\theta}\, d\theta = -\pi i.$

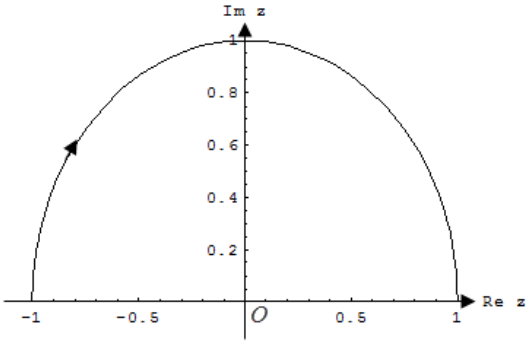


Fig: 8

Exercise Set: 6.1

1. Sketch the path C from $z=0$ to $z=3i$ and hence evaluate $\int_C z^2\,dz$.
2. Sketch the path C from $z=0$ to $z=3$ and hence evaluate $\int_C \bar{z}\,dz$.
3. Sketch the path C from $z=1$ to $z=4$ and hence evaluate $\int_C \sin z\,dz$.
4. Sketch the path C, which is the unit circle $|z|=2$ and hence evaluate $\int_C (z+\bar{z})dz$.

Sketch the corresponding paths and hence evaluate them (5-8):

5. $\int_C (e^{2z}+\cos z)\,dz$, C is the shortest path from $z=2$ to $z=4$.
6. $\int_C (z\,.\,\bar{z})\,dz$, C is the path around the square with vertices $0,1,1+i,i$.
7. $\int_C \left(\frac{1}{z-i} - \frac{2}{(z-i)^2}\right)dz$, C is the circle $|z-i|=3$, clockwise.
8. $\int_C (x+y-i\,x^2)\,dz$, C is the shortest path along the real axis from $z=0$ to $z=1$ and then along a line parallel to imaginary axis from $z=1$ to $z=1+i$.

9. **Problems From Book: (**Ref: Dennis Zill - A first course in complex analysis with applications-Jones and Bartlett (2003); Page# 254-256**)**

In the following problems, evaluate the given integral along the indicated contour.

a. $\int_C (z+3)dz$, where C is $x=2t, y=4t-1, 1\le t\le 3$

b. $\int_C z^2 dz$, where C is $z(t)=3t+2it,\ -2\le t\le 2$

c. $\int_C \frac{z+1}{z}dz$, where C is the right half of the circle $|z|=1$ from $z=-i$ to $z=i$

d. $\oint_C Re(z)dz$, where C is the circle $|z| = 1$

e. $\oint_C \left(\frac{1}{(z+i)^3} - \frac{5}{z+i} + 8\right) dz$, where C is the circle $|z+i| = 1, 0 \le t \le 2\pi$

f. $\int_C (x^2 + iy^3)dz$, where C is the straight line from $z = 1$ to $z = i$

g. $\int_c (x^2 - iy^3)\, dz$, where C is the lower half of the circle $|z| = 1$ from $z = -1$ to $z = 1$

h. $\int_C e^z dz$, where C is the polygonal path consisting of the line segments from $z = 0$ to $z = 2$ and from $z = 2$ to $z = 1 + \pi i$.

i. $\int_C \sin z\, dz$, where C is the polygonal path consisting of the line segments from $z = 0$ to $z = 1$ and from $z = 1$ to $z = 1 + i$

j. $\int_C Im(z - i)\, dz$, where C is the polygonal path consisting of the circular arc along $|z| = 1$ from $z = 1$ to $z = i$ and the line segment from $z = i$ to $z = -1$

k. $\int_C dz$, where C is the left half of the circle $|z| = 2$ from $z = 2i$ to $z = -2i$.

l. $\oint_C z\, e^z\, dz$, where C is the square with the vertices $z = 0, z = 1, z = 1 + i$, and $z = i$.